

Core Fundamental Windows CMD Commands Guide

This document contains essential and fundamental Windows Command Prompt (CMD) commands that every developer, engineer, and power user should know.

File & Folder Navigation

dir

Lists files and folders in the current directory.

```
dir
dir /w
```

cd

Changes directory.

```
cd foldername
cd ..
cd \
```

mkdir / md

Creates a new folder.

```
mkdir Projects
md Projects
```

rmdir / rd

Deletes a folder.

```
rmdir foldername
rmdir /s foldername
```

File Operations

copy

Copies files.

```
copy file.txt backup.txt
```

move

Moves files.

```
move file.txt Documents
```

del

Deletes files.

```
del file.txt
del *.txt
```

rename / ren

Renames files.

```
ren old.txt new.txt
```

type

Displays file contents.

```
type notes.txt
```

System Information

cls

Clears the terminal.

```
cls
```

echo

Prints text or writes into files.

```
echo Hello World  
echo Hello > file.txt
```

systeminfo

Shows detailed system information.

```
systeminfo
```

hostname

Shows computer name.

```
hostname
```

tasklist

Shows running processes.

```
tasklist
```

taskkill

Kills a process.

```
taskkill /IM chrome.exe /F
```

Networking Commands

ipconfig

Shows IP and network information.

```
ipconfig
```

```
ipconfig /all
```

ping

Tests internet/server connection.

```
ping google.com
```

tracert

Shows network route.

```
tracert google.com
```

netstat

Shows active network connections.

```
netstat  
netstat -ano
```

Disk & Storage

tree

Displays folder structure visually.

```
tree  
tree /f
```

diskpart

Advanced disk management tool.

```
diskpart
```

chkdsk

Checks disk health.

```
chkdsk
```

Permissions & Admin

runas

Runs program as administrator or another user.

```
runas /user:Administrator cmd
```

sfc

Repairs corrupted Windows system files.

```
sfc /scannow
```

Useful Developer Commands

where

Finds executable location.

```
where python
```

path

Shows PATH environment variable.

```
path
```

set

Shows environment variables.

```
set
```

help

Shows all commands or help for specific commands.

```
help  
help dir
```

Extremely Useful Shortcuts

Shortcut	Meaning
Tab	Autocomplete
↑	Previous command
↓	Next command
Ctrl + C	Stop command
cls	Clear screen

Commands Every Engineer Should Memorize

```
dir  
cd  
mkdir  
rmdir  
copy  
move  
del  
echo  
type  
ipconfig  
ping  
tasklist  
taskkill  
where  
help
```

Example Mini Workflow

```
mkdir AI_Project  
cd AI_Project  
echo print("hello") > main.py
```

```
dir
type main.py
```

Next Level After CMD

- PowerShell
- Linux terminal/bash
- Git terminal
- Docker CLI
- SSH
- Python scripting
- Automation with batch files